

PMP Certification - Key Concepts

1. INTRODUCTION TO PROJECT MANAGEMENT

What is a Project?

- A temporary endeavour undertaken to create a unique product, service or result. "Temporary" refers to the project itself and the team, not to the deliverable, which may last for years after the project ends.
- Has a definite start and end date. A project ends when its objectives are achieved, when they cannot be met, when funding runs out, when the need no longer exists, or when it is terminated for legal/convenience reasons.
- Example: Building a bridge is a project - once the bridge is built and handed over, the project ends, but the bridge itself continues to exist and be used (that is operations).

What is Project Management?

- Application of knowledge, skills, tools and techniques to project activities to meet project requirements. It is how a PM turns a plan into a completed deliverable.
- Involves balancing the competing demands of scope, schedule, cost, quality, resources and risk, while satisfying different stakeholder needs and expectations.
- It is achieved through appropriate application of the process groups (Initiating, Planning, Executing, Monitoring & Controlling, Closing) and tailoring the approach to the project's context.

Temporary vs Ongoing Operations

- Project = temporary, unique, creates a specific output and then closes; example: developing a new mobile app.
- Operations = ongoing, repetitive, sustains the business over time; example: running the customer support desk after the app is launched.
- Note: recurring, repetitive work with no defined end describes operations, not a project.

Project Characteristics

- Temporary: has a clear beginning and end.
- Produces a unique output: even similar projects (e.g. building two houses) differ in some way (location, buyer requirements).
- Progressively elaborated: details are developed incrementally as more information becomes available, while overall scope stays controlled.
- Has a defined objective/purpose tied to organizational strategy.
- Consumes resources (people, money, materials, time) and involves inherent uncertainty and risk.

Project Constraints

Triple Constraint

- Scope, Time (Schedule) and Cost - the three classic limiting factors of any project. They are interrelated: changing one usually forces a change in at least one of the others (e.g. adding scope usually increases cost and/or time).

Expanded Constraints

- Modern PM also recognises Quality, Resources and Risk as constraints that must be balanced along with scope, time and cost. All six constraints must be considered together when planning and when evaluating a proposed change.

Project Success Criteria

- Traditionally measured by meeting scope, time, cost and quality targets (the "iron triangle" plus quality).
- Modern view also includes stakeholder satisfaction, achievement of business value/benefits, and whether the project delivered what the organization actually needed - a project can be "on time, on budget" and still be considered unsuccessful if it fails to deliver value.

Project Life Cycle

- The series of phases a project passes through from start to finish: Starting the project, Organizing and Preparing, Carrying out the work, and Closing the project.
- Can follow different development approaches: Predictive (fully planned upfront), Iterative (repeated cycles refining the product), Incremental (deliverables produced in usable increments), Adaptive/Agile (short iterations responding to change), or Hybrid (a mix).

Product Life Cycle

- The stages a product goes through over its commercial life: Introduction, Growth, Maturity and Decline/Retirement.
- A single product life cycle can span multiple project life cycles - for example, a software product may have separate projects for its initial launch, for major version 2.0, and later for end-of-life migration.

Program vs Portfolio vs Project

- Project: temporary work producing a unique product, service or result.
- Program: a group of related projects, subsidiary programs and program activities managed in a coordinated way to obtain benefits not available from managing them individually (e.g. an "airport expansion program" containing runway, terminal, and parking projects).
- Portfolio: a collection of projects, programs, sub-portfolios and operations managed together as a group to achieve strategic business objectives; portfolio components may not be related to each other, unlike program components.

Project Management Office (PMO)

- An organizational structure/department that standardizes project governance processes and facilitates sharing of resources, methodologies, tools and techniques across projects.

Types of PMO

- Supportive PMO: plays a consultative role, supplies templates, best practices, training and access to information; low degree of control.
- Controlling PMO: provides support and requires compliance with frameworks, methodologies, templates or governance; moderate degree of control.
- Directive PMO: takes direct control by actually managing the projects themselves; high degree of control.

Organizational Structures

- Functional: staff grouped by department/specialty (e.g. marketing, IT); the PM has little to no formal authority and the functional manager controls the budget and resources.
- Matrix: a blend of functional and projectized structures where staff report to both a functional manager and a PM.
- Weak Matrix: keeps most characteristics of a functional org; the PM acts more like a coordinator/expediter with limited authority.
- Balanced Matrix: PM has moderate authority; power is shared fairly evenly between PM and functional manager.
- Strong Matrix: PM has significant authority and generally works on projects full time, similar to a projectized org.
- Projectized: the entire organization is structured around projects; the PM has high to almost total authority and team members report directly to the PM; once the project ends, staff may need to find their next assignment.

Project Manager Responsibilities

- Leading and directing the project team toward achieving the project objectives within the agreed constraints.
- Developing and maintaining the project management plan, managing risk, resolving issues, ensuring quality, and integrating all the moving parts of the project.
- Communicating with and managing expectations of stakeholders, and escalating decisions that are outside the PM's authority.

Leadership vs Management

- Leadership: inspiring, motivating and influencing people; setting a vision and direction; often more about "why".
- Management: planning, organizing, directing and controlling resources/processes to reach defined goals; more about "how".
- Effective PMs blend both - management skills keep the project on track day to day, while leadership skills are essential for guiding teams through change, conflict, and uncertainty.

2. BUSINESS ENVIRONMENT

Business Value

- The net quantifiable benefit (tangible and intangible) that results from a business endeavour.
- Tangible value: monetary assets, fixtures, stockholder equity, market share.
- Intangible value: goodwill, brand recognition, public benefit, reputation, strategic alignment - harder to measure but equally important when justifying a project.

Benefits Management

- The process of identifying, defining, planning, tracking and realizing the benefits a project is expected to deliver.

- Documented in the Benefits Management Plan, created during Initiating, and monitored throughout and even after the project (since some benefits are only realized after the deliverable is in use).

Business Case

- A documented economic feasibility study used to establish the validity of the benefits of a selected project component that lacks sufficient definition.
- Used before the Project Charter to justify investment; includes business need, situation analysis, options considered, cost-benefit analysis, and recommendation.

Strategic Alignment

- Ensuring the project's objectives directly support the organization's strategic goals, mission and priorities.
- Projects that are not strategically aligned risk wasting resources and are more likely to be cancelled during portfolio reviews.

Value Delivery System

- The collection of strategic business activities aimed at building, sustaining and/or advancing an organization; it includes portfolios, programs, projects, products and operations working together as a system to create value for stakeholders.
- Emphasizes that projects don't exist in isolation - they are one component that helps convert organizational strategy into business value.

Governance

- The framework of authority, decision-making and oversight that guides an organization's overall activities, including how project decisions are made, who approves them, and how accountability is enforced.
- Project governance typically includes a sponsor, steering committee, and defined escalation paths.

Compliance

- Adhering to laws, regulations, industry standards, contracts and internal organizational policies relevant to the project.
- Non-compliance can create legal, financial and reputational risk, so compliance requirements should be identified early and tracked as constraints.

Organizational Change Management

- A structured, deliberate approach for transitioning individuals, teams and the organization from a current state to a desired future state.
- Involves managing the people side of change: communication, training, stakeholder buy-in, and reinforcement, so the project's deliverables are actually adopted and benefits realized.

Organizational Process Assets (OPA)

- Internal plans, processes, policies, procedures, templates and historical knowledge (lessons learned, historical information) that the organization has built up and that influence how the project is managed.
- Used as inputs throughout planning to avoid reinventing the wheel and to leverage past experience.

Enterprise Environmental Factors (EEF)

- Conditions, internal or external to the organization, that are not under the direct control of the project team but influence, constrain or direct the project.
- Examples: organizational culture, market conditions, government regulations, infrastructure, existing HR policies, and industry standards.

3. PROJECT MANAGEMENT PRINCIPLES (12 PRINCIPLES)

- Stewardship: Be a diligent, respectful and caring steward of resources; act responsibly, ethically, with integrity, and be mindful of financial, human, material and environmental resources entrusted to the project.
- Team: Build a culture of team accountability and respect; create a collaborative project team environment where people feel able to contribute their best.
- Stakeholders: Engage stakeholders proactively and to the degree needed to understand their interests, needs and expectations, rather than waiting for them to raise concerns.
- Value: Focus continuously on value - the ultimate indicator of project success - by re-evaluating alignment with intended business outcomes throughout the project, not just at the end.
- Systems Thinking: Recognize, evaluate and respond to the dynamic circumstances within and around the project holistically; understand how changes in one part of the system affect other parts.
- Leadership: Demonstrate and adapt leadership behaviours to support individual and team needs; leadership can come from anyone on the team, not only the PM.
- Tailoring: Tailor the project approach based on the project's context, size, complexity and organizational environment - there is no single method that fits every project.
- Quality: Build quality into processes and deliverables from the start, focusing on meeting acceptance criteria and stakeholder needs rather than inspecting quality in at the end.
- Complexity: Navigate complexity (arising from human behaviour, system behaviour, ambiguity and technological innovation) by using experience, learning, and iterative approaches rather than trying to plan away all uncertainty.
- Risk: Optimize risk responses by proactively addressing both threats and opportunities to maximize positive impact and minimize negative impact on objectives.
- Adaptability and Resilience: Build adaptability and resilience into the project approach so the team can respond effectively to change and recover quickly from setbacks or failures.
- Change: Enable change to achieve the envisioned future state; prepare those affected by the change (people, processes) for adoption and long-term sustainment of the change.

4. PERFORMANCE DOMAINS (8 DOMAINS)

1. Stakeholder Performance Domain

- Identification: the ongoing process of finding all individuals, groups or organizations who may affect, be affected by, or perceive themselves to be affected by the project - done continuously, not just once.
- Analysis: assessing each stakeholder's power, interest, influence, and attitude toward the project, commonly using tools like the Power/Interest grid to decide engagement strategy.

- Engagement: actively working with stakeholders to understand and meet their needs/expectations, and to address issues as they arise, matching the Stakeholder Engagement Plan.
- Communication: keeping stakeholders informed at the right level of detail, frequency and format for their role and interest.

2. Team Performance Domain

- Team development: teams typically move through Forming (getting acquainted), Storming (conflict as roles are worked out), Norming (working relationships stabilize), Performing (high productivity) and Adjourning (team disbands) - known as the Tuckman ladder.
- Motivation: understanding what drives performance - intrinsic motivation (personal satisfaction, growth) generally sustains performance better than extrinsic motivation (rewards, bonuses) alone; theories include Maslow's hierarchy of needs, Herzberg's hygiene/motivator factors, and McGregor's Theory X/Y.
- Conflict management: resolving disagreements constructively using techniques such as collaborate/problem-solve, compromise, smooth/accommodate, force/direct, or withdraw/avoid, chosen based on urgency and relationship importance.
- Emotional intelligence: the capability to identify and manage one's own emotions and the emotions of others; includes self-awareness, self-management, social awareness and relationship management.
- Servant leadership: a leadership style where the PM's primary role is to serve the team by removing obstacles, providing resources, and empowering people rather than directing every action.

3. Development Approach and Life Cycle Performance Domain

- Predictive approach: requirements, scope, schedule and cost are planned in detail up front and executed largely as planned; best when requirements are well understood and stable (e.g. Waterfall).
- Agile approach: work is delivered iteratively and incrementally, embracing changing requirements throughout the project; best when requirements are uncertain or likely to evolve.
- Hybrid approach: combines predictive and agile elements - for example predictive planning/governance with agile execution - used when parts of the project are well defined and parts are not.
- Choosing the right approach depends on factors such as: how well requirements are understood, product/technology variability, need for frequent stakeholder feedback, delivery cadence required, safety/regulatory constraints, and organizational culture.

4. Planning Performance Domain

- Planning levels: planning happens at multiple levels - strategic/portfolio level, program level, and project level (which itself has phase, iteration and daily/detailed planning).
- Rolling wave planning: near-term work is planned in detail while work further in the future is planned only at a high level, and is elaborated as it gets closer - useful when full detail isn't yet known.
- Progressive elaboration: continuously improving and adding more detail to a plan as more accurate and complete information becomes available as the project progresses; related to, but broader than, rolling wave planning.

5. Project Work Performance Domain

- Managing work: coordinating people, processes and resources to keep tasks moving and remove obstacles/blockers as they surface.
- Deliverables: any unique and verifiable product, result or capability that must be produced to complete a process, phase or project.
- Quality: building processes that ensure deliverables meet the defined requirements and acceptance criteria before being handed over.
- Procurement: acquiring the goods, services or results needed from outside the performing team when internal capability/capacity is insufficient.
- Delivery: the act of providing the completed and accepted deliverables/outcomes to the customer or end user.

6. Delivery Performance Domain

- Value delivery: ensuring the project actually produces the intended business value and benefits described in the business case, not just the contracted deliverables.
- Customer satisfaction: meeting or exceeding customer expectations by understanding their true needs, involving them throughout, and managing their expectations proactively.
- Product acceptance: the formal process by which the customer or sponsor reviews and signs off that deliverables meet the agreed acceptance criteria.

7. Measurement Performance Domain

- KPIs and Metrics: quantifiable measures used to track project health and performance against baselines (e.g. cost variance, defect rate, on-time delivery percentage).
- Earned Value Management (EVM): integrates scope, schedule and cost data to objectively measure project performance and forecast future results (see Section 15 for full detail).
- Performance reporting: communicating current project status, progress and forecasts to stakeholders through reports, dashboards, and status meetings, tailored to audience needs.

8. Uncertainty Performance Domain

- Risk: an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives.
- Opportunity: a risk that would have a positive effect on one or more project objectives if it occurs - opportunities should be actively pursued, not just accepted.
- Resilience: the project team's ability to absorb the impact of a disruptive event and recover quickly, adapting the plan as needed.
- Ambiguity: a state of being unclear about facts, intent or how to proceed, often addressed through experiments, prototypes, spikes and incremental steps rather than trying to plan it all away up front.

5. PREDICTIVE (WATERFALL) APPROACH

Initiating Process Group

- Business Case: justifies why the project should be undertaken from a financial/strategic point of view; created before the charter.

- **Benefits Management Plan:** describes how and when the project's intended benefits will be delivered and how they will be measured.
- **Project Charter:** the document that formally authorizes the existence of the project and gives the PM authority to apply organizational resources; includes high-level objectives, assumptions, constraints, and the named sponsor/PM.
- **Stakeholder Register:** an initial list of stakeholders, including their identification, assessment and classification information (power/interest, expectations, requirements).

Planning Process Group

Scope

- **Scope Management Plan:** describes how scope will be defined, developed, monitored, controlled and validated throughout the project.
- **Requirements:** documented needs and expectations of stakeholders that the project/product must satisfy, gathered through interviews, workshops, surveys, prototypes, etc.
- **Requirements Traceability Matrix:** a table that links each requirement to its origin (business need) and tracks it throughout the project life cycle to ensure nothing is lost and everything delivered is actually needed.
- **WBS (Work Breakdown Structure):** a hierarchical decomposition of the total scope of work into smaller, more manageable components called work packages.
- **WBS Dictionary:** provides detailed information about each WBS component - description, acceptance criteria, resources needed, cost estimate.
- **Scope Baseline:** the approved version of the Scope Statement, WBS, and WBS Dictionary; can be changed only through formal change control.
- **Scope Validation:** the formal process of obtaining the sponsor's/customer's acceptance of completed project deliverables (focuses on acceptance, done during Monitoring & Controlling).
- **Scope Control:** monitoring the status of the project and product scope and managing changes to the scope baseline, including identifying and controlling scope creep.

Schedule

- **Activity List:** a documented list of all schedule activities required to complete the project, with enough detail for team members to understand the work required.
- **Milestones:** significant points or events in the project with zero duration, used to mark progress (e.g. "Design approved").
- **Network Diagram:** a visual/graphical representation of the logical relationships (sequence) among project schedule activities.
- **Dependencies:** Mandatory (hard logic, legally/contractually required), Discretionary (soft logic, preferred sequence based on best practice), External (relationship between project and non-project activities), Internal (relationship between project activities, generally within the team's control).
- **Critical Path:** the longest sequence of activities through the network diagram, determining the shortest possible project duration; activities on this path have zero (or the least) float and any delay here delays the whole project.
- **Float (Slack):** the amount of time an activity can be delayed without delaying the project end date (Total Float) or without delaying the early start of a successor activity (Free Float).

- Lead and Lag: Lead = allows an activity to start before its predecessor finishes (overlap, shown as negative time); Lag = imposes a waiting period before a successor can start (e.g. curing concrete).
- Gantt Chart: a bar chart displaying schedule activities against a timeline, useful for communicating schedule status to stakeholders.
- Schedule Baseline: the approved version of the schedule model used as the basis for comparing actual results.
- Crashing: a schedule compression technique that adds more resources to critical path activities to shorten duration - this increases cost and doesn't always work (adding people to a late task can even slow it down).
- Fast Tracking: a schedule compression technique where activities normally done in sequence are performed in parallel, at least partially - this increases risk of rework.

Cost

- Cost Estimation: predicting the likely cost of the resources needed to complete project activities, using techniques such as analogous (uses historical data, fast but less accurate), parametric (uses a statistical relationship), bottom-up (estimates each work package and rolls up, most accurate but time-consuming), and three-point/PERT estimating.
- Budget: the sum of the approved cost estimates for the project, aggregated at the work package and control account levels to form the overall project budget.
- Cost Baseline: the approved, time-phased budget used as a basis against which to measure and monitor actual cost performance.
- Contingency Reserve: budget set aside for identified risks that are accepted, part of the cost baseline, and can be used at the PM's discretion for those "known-unknowns".
- Management Reserve: budget set aside for unforeseen work within scope (the "unknown-unknowns"); NOT part of the cost baseline and requires formal change control approval to access.

Quality

- Quality Planning: identifying quality requirements and/or standards for the project and its deliverables, and documenting how the project will demonstrate compliance.
- Quality Assurance (Manage Quality): the process of auditing quality requirements and results to ensure appropriate quality standards and operational definitions are used - focuses on process improvement.
- Quality Control (Control Quality): monitoring and recording the results of executing quality activities to assess performance and recommend necessary changes - focuses on the correctness of the deliverable itself.
- Cost of Quality: Cost of Conformance (prevention costs - training, documenting processes; plus appraisal costs - testing, inspections) versus Cost of Nonconformance (internal failure costs - rework, scrap; plus external failure costs - warranty claims, lost business).
- Seven Basic Quality Tools: Cause-and-effect (fishbone/Ishikawa) diagrams, Flowcharts, Checksheets, Pareto diagrams, Histograms, Control charts, and Scatter diagrams - used to identify, analyze and resolve quality issues.

Resources

- Resource Planning: identifying and documenting the roles, responsibilities, required skills, and reporting relationships needed for the project, and determining how many/what type of resources are needed.

- Responsibility Assignment Matrix (RACI): a matrix showing who is Responsible (does the work), Accountable (owns the outcome, usually one person), Consulted (gives input), and Informed (kept up to date) for each activity or deliverable.
- Resource Histogram: a bar chart showing the amount of resource time/quantity needed or used across the project timeline, helping spot over-allocation.
- Team Charter: a document that establishes the team's values, agreements and operating guidelines (e.g. working hours, communication norms, decision-making process) to promote a healthy team culture.

Communications

- Communication Planning: determining stakeholders' information needs (what, when, how, by whom) and documenting the approach in the Communications Management Plan.
- Communication Channels Formula: $N(N-1)/2$, where N is the number of stakeholders/team members - shows how quickly communication complexity grows as the team gets bigger.
- Information Distribution: the process of making relevant information available to project stakeholders in a timely manner, through the agreed channels and formats.
- Reports: status reports, progress reports and dashboards that summarize project performance data for different audiences.

Risk

- Risk Register: the central log recording identified risks, their category, root cause, triggers, owner, probability/impact, and planned responses; updated throughout the project.
- Risk Report: a document that presents information on overall (aggregate) project risk exposure and summarizes the most important individual project risks.
- Risk Identification: the ongoing process of determining which risks might affect the project, using techniques such as brainstorming, checklists, SWOT analysis, assumption analysis and expert judgement.
- Qualitative Risk Analysis: prioritizing risks for further analysis by assessing and combining their probability of occurrence and impact, typically using a Probability and Impact Matrix.
- Quantitative Risk Analysis: numerically analyzing the combined effect of identified risks on overall project objectives, using techniques like Monte Carlo simulation, decision tree analysis, or Expected Monetary Value (EMV).
- Risk Responses: for Threats - Avoid (eliminate the threat), Mitigate (reduce probability/impact), Transfer (shift ownership, e.g. insurance), Accept (do nothing or set aside reserve), Escalate (pass to a higher level if outside project authority). For Opportunities - Exploit (ensure it happens), Enhance (increase probability/impact), Share (partner with a third party to capture it), Accept, Escalate.
- Residual Risk: the risk that remains after a planned risk response has been implemented; must still be monitored.
- Secondary Risk: a new risk that arises directly as a result of implementing a risk response to another risk.

Procurement

- Contract Types: see the dedicated Procurement section below for full detail on Fixed Price, Cost Reimbursable, and Time & Material contracts.
- Procurement Documents: RFP (Request for Proposal - detailed solution needed, price is one of several factors), RFQ (Request for Quote - price is the main factor, requirements are well defined), RFI (Request for Information - used early, to gather general information about potential sellers/solutions).

- **Bid Evaluation:** comparing seller proposals against pre-established weighted evaluation criteria (technical capability, price, past performance, delivery schedule) to select the best-fit seller.
- **Procurement Strategy:** the overall approach for how the project's procurements will be conducted, including delivery method, contract payment type, and how procurement phases will be sequenced.

Stakeholder

- **Stakeholder Engagement Plan:** documents the strategies and specific actions required to promote productive stakeholder involvement, based on their needs, interests and potential impact on the project.

Executing Process Group

- **Direct and Manage Project Work:** leading and performing the work defined in the project management plan and implementing approved changes to achieve the project's objectives.
- **Manage Quality:** translating the quality management plan into executable quality activities, including audits, and identifying process improvements.
- **Acquire Resources:** obtaining team members, facilities, equipment, materials and supplies needed to complete the project work.
- **Develop Team:** improving team members' competencies, team member interaction, and overall team environment to enhance performance.
- **Manage Team:** tracking team member performance, providing feedback, resolving issues and managing team changes to optimize project performance.
- **Manage Communications:** creating, collecting, distributing, storing, retrieving and disposing of project information appropriately according to the communications plan.
- **Implement Risk Responses:** executing the agreed-upon risk response plans at the appropriate time so identified risks are actually addressed, not just documented.
- **Conduct Procurements:** obtaining seller responses, selecting a seller, and awarding a contract based on the evaluation criteria.
- **Manage Stakeholder Engagement:** communicating and working with stakeholders to meet their needs/expectations, address issues, and foster appropriate engagement.

Monitoring & Controlling Process Group

- **Integrated Change Control:** reviewing all change requests, approving or rejecting them (often via a Change Control Board), and managing changes to deliverables, project documents and the project management plan.
- **Performance Measurement:** tracking actual performance against the planned baselines using metrics and analysis techniques.
- **Variance Analysis:** comparing planned versus actual results to determine the size and cause of any variance and whether corrective action is needed.
- **Scope Control:** monitoring the status of scope and managing changes to the scope baseline, guarding against scope creep.
- **Schedule Control:** monitoring the status of project activities to update project progress and manage changes to the schedule baseline.
- **Cost Control:** monitoring the status of the project to update the project budget and manage changes to the cost baseline.

- **Risk Monitoring:** tracking identified risks, monitoring residual risks, identifying new risks, and evaluating the effectiveness of the risk process throughout the project.
- **Procurement Monitoring:** managing procurement relationships, monitoring contract performance, and making changes/corrections to contracts as needed.
- **Earned Value:** a technique that objectively measures project performance and progress by comparing planned value, earned value and actual cost (see Section 15).

Closing Process Group

- **Project Closure:** finalizing all activities across all process groups to formally complete the project, phase, or contract.
- **Lessons Learned:** knowledge gained during the project - what went well, what didn't, and recommendations - documented for the benefit of future projects.
- **Final Report:** a summary of overall project performance (scope, schedule, cost, quality, benefits achieved) shared with stakeholders at closure.
- **Handover:** formally transferring the final product, service or result to the operations team, maintenance team, or customer for ongoing use.
- **Contract Closure:** verifying that all contractual obligations were completed satisfactorily, resolving any open claims, and formally closing out procurement agreements.

6. KNOWLEDGE AREAS (10)

For each knowledge area, know the general purpose along with typical high-level inputs (plans, documents), tools & techniques (analysis methods, meetings, software), and outputs (updated plans/registers):

- **Integration Management:** unifies, consolidates, communicates and integrates all the other knowledge areas; includes developing the Charter and PM Plan, directing project work, managing knowledge, monitoring work, performing integrated change control, and closing the project/phase.
- **Scope Management:** defines and controls what work is, and is not, included in the project (planning scope, collecting requirements, defining scope, creating WBS, validating and controlling scope).
- **Schedule Management:** plans, develops, manages, executes and controls the schedule to ensure timely completion of the project.
- **Cost Management:** plans, estimates, budgets, finances and controls costs so the project can be completed within the approved budget.
- **Quality Management:** plans, manages and controls quality requirements/standards so the project satisfies the needs it was undertaken for.
- **Resource Management:** identifies, acquires and manages the physical and team resources needed for successful project completion.
- **Communications Management:** ensures timely and appropriate planning, collection, creation, distribution, storage, retrieval and disposition of project information.
- **Risk Management:** plans, identifies, analyzes, plans responses to, implements responses to, and monitors risk on a project.
- **Procurement Management:** plans, conducts, controls and closes the purchasing/acquiring of products, services or results needed from outside the project team.

- Stakeholder Management: identifies people/groups/organizations impacted by the project, analyzes their expectations, and develops strategies to effectively engage them in project decisions and execution.

7. AGILE

Agile Values (Agile Manifesto)

- Individuals and interactions over processes and tools - people and how they work together matter more than rigid procedures.
- Working software over comprehensive documentation - a usable product proves progress better than paperwork, though some documentation is still needed.
- Customer collaboration over contract negotiation - ongoing partnership with the customer beats rigidly following an initial contract.
- Responding to change over following a plan - the ability to adapt is valued more than sticking to an original plan when circumstances change.
- Note: the manifesto values the items on the right too, just less than the items on the left.

12 Agile Principles

- Highest priority is to satisfy the customer through early and continuous delivery of valuable software/product.
- Welcome changing requirements, even late in development - agile processes harness change for the customer's competitive advantage.
- Deliver working product frequently, from a couple of weeks to a couple of months, with a preference for shorter timescales.
- Business people and developers must work together daily throughout the project.
- Build projects around motivated individuals; give them the environment, support and trust they need to get the job done.
- Face-to-face conversation is the most efficient and effective method of conveying information within a team.
- Working software/product is the primary measure of progress, not documents or plans.
- Agile processes promote sustainable development - sponsors, developers and users should be able to maintain a constant pace indefinitely.
- Continuous attention to technical excellence and good design enhances agility.
- Simplicity - maximizing the amount of work not done - is essential.
- The best architectures, requirements and designs emerge from self-organizing teams.
- At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behaviour accordingly (retrospectives).

Agile Frameworks

- Scrum: an iterative, incremental framework using fixed-length sprints, defined roles (Product Owner, Scrum Master, Developers), artifacts (Backlogs, Increment), and events (Sprint Planning, Daily Scrum, Review, Retrospective).

- Kanban: a visual workflow method that limits work-in-progress (WIP) and uses a continuous pull system rather than fixed-length iterations; focuses on flow and cycle time.
- Lean: focuses on eliminating waste, amplifying learning, deciding as late as possible, delivering as fast as possible, and building quality/integrity into the product from the start.
- XP (Extreme Programming): emphasizes engineering practices such as pair programming, test-driven development, continuous integration, and simple design to improve software quality and responsiveness.
- Crystal: a family of methodologies (Crystal Clear, Yellow, Orange, etc.) tailored by team size and project criticality; emphasizes people, interaction, community and communication over rigid process.
- DSDM (Dynamic Systems Development Method): fixes time, cost and quality up front while varying features/scope; uses timeboxing and MoSCoW prioritization (Must have, Should have, Could have, Won't have).
- FDD (Feature-Driven Development): an iterative, model-driven approach that organizes development around building/designing features, planned and tracked feature by feature.

Scrum Roles

- Product Owner: owns and prioritizes the Product Backlog to maximize the value of the product; represents the voice of the customer/stakeholders and makes decisions about what gets built.
- Scrum Master: a servant leader who coaches the team, removes impediments, facilitates Scrum events, and ensures the Scrum framework is understood and followed by the team and organization.
- Developers: the cross-functional group of people who do the actual work of building a "done" increment each sprint; self-organizing and collectively responsible for the work.

Scrum Artifacts

- Product Backlog: an ordered, evolving list of everything that might be needed in the product, prioritized by the Product Owner based on value.
- Sprint Backlog: the subset of Product Backlog items selected for a given sprint, plus a plan for delivering the increment and achieving the sprint goal.
- Increment: the sum of all Product Backlog items completed during a sprint and all previous sprints, which must meet the team's Definition of Done and be potentially releasable.

Scrum Events

- Sprint: a fixed-length timebox, usually 1-4 weeks, during which a "done", usable increment is created; contains all the other events.
- Sprint Planning: the team plans the work to be done during the upcoming sprint, defining the sprint goal and selecting backlog items.
- Daily Scrum: a short (typically 15-minute) daily meeting where the developers synchronize activities and plan for the next 24 hours.
- Sprint Review: held at the end of the sprint to inspect the increment and gather feedback from stakeholders, informally adapting the product backlog if needed.
- Sprint Retrospective: the team reflects on the past sprint regarding people, relationships, process and tools, and identifies concrete improvements for the next sprint.

Agile Estimation

- Story Points: a relative measure of the effort, complexity and uncertainty of a backlog item, not tied directly to hours; helps teams estimate faster and more consistently over time.
- Planning Poker: a consensus-based estimation technique where team members privately select an estimate card (often Fibonacci-like numbers) and reveal simultaneously, discussing outliers until consensus is reached.
- Velocity: the average amount of work (usually measured in story points) a team completes per sprint, used to forecast how much work can be done in future sprints.
- Burn-down Chart: shows the amount of work remaining over time within a sprint or release; ideally trends down to zero by the end.
- Burn-up Chart: shows both the amount of work completed and the total project scope over time (two separate lines), making scope changes easy to see.

Agile Leadership

- Servant Leadership, Coaching, Facilitation and Empowerment: the Scrum Master/agile leader supports the team by removing blockers, teaching skills, guiding discussions and giving the team autonomy to make decisions, rather than issuing top-down directives.

Agile Quality

- Definition of Done (DoD): the team's shared, explicit understanding of what "complete" means for a backlog item or increment (e.g. coded, tested, documented, reviewed) - ensures consistent quality.
- Definition of Ready (DoR): the criteria a backlog item must meet before it can be pulled into a sprint (e.g. clear acceptance criteria, estimated, dependencies identified).
- Continuous Integration: the practice of frequently merging code changes into a shared repository, with automated builds/tests run on each merge to catch integration issues early.
- Test Driven Development (TDD): a practice where developers write an automated test before writing the code that makes the test pass, driving simpler and more reliable design.

8. HYBRID PROJECT MANAGEMENT

- When to use Hybrid: appropriate when part of a project has stable, well-understood requirements suited to predictive planning (e.g. procurement, regulatory documentation) while another part is uncertain or evolving and benefits from agile iteration (e.g. new software feature development) - common in construction-plus-technology or product-plus-software projects.
- Combining Agile and Predictive: common patterns include predictive high-level scope/budget/milestone planning with agile sprints for execution, or "agile-fall" where requirements/design use predictive methods and build/test use agile sprints.
- Governance in Hybrid projects: still requires clear reporting lines, an integrated change control process, and blended metrics (e.g. combining EVM for the predictive parts with velocity/burndown for the agile parts) so stakeholders get a consistent view of overall progress.

9. LEADERSHIP & SOFT SKILLS

- Emotional Intelligence: the ability to identify, understand and manage your own emotions and recognize/influence the emotions of others; critical for building trust and leading diverse teams.

- **Conflict Resolution:** choose the technique based on the situation - Collaborate/Problem-solve (win-win, best for important/complex issues), Compromise (both sides give up something, lose-lose), Smooth/Accommodate (preserve relationship, minor issues), Force/Direct (win-lose, urgent situations), Withdraw/Avoid (postpone, low-priority issues).
- **Negotiation:** the process of reaching a mutually acceptable agreement; principled negotiation focuses on interests (the "why") rather than fixed positions (the "what"), aiming for win-win outcomes.
- **Motivation:** understanding intrinsic drivers (personal satisfaction, mastery, autonomy, purpose) versus extrinsic drivers (pay, bonuses, recognition) helps a PM tailor how to keep the team engaged; relevant models include Maslow's Hierarchy of Needs and Herzberg's Two-Factor (hygiene/motivator) theory.
- **Coaching:** a supportive, question-based approach that helps an individual find their own solutions and improve performance/skills over time.
- **Mentoring:** a more experienced person shares knowledge and guides the career/professional growth of a less experienced person, typically over a longer relationship than coaching.
- **Active Listening:** fully concentrating on, understanding, and thoughtfully responding to what is being said (verbally and non-verbally) rather than just waiting to speak.
- **Decision Making:** PMs can use autocratic (PM decides alone, fast but low buy-in), consultative (PM decides after gathering input), or consensus-based (group agrees together, slower but high buy-in) approaches depending on urgency and stakes.
- **Facilitation:** guiding a group discussion or workshop toward an effective outcome while ensuring all voices are heard and the group stays on track.
- **Critical Thinking:** objectively analyzing information, questioning assumptions, and evaluating evidence before forming a judgement or making a decision.
- **Problem Solving:** a structured approach - define the problem, identify the root cause (not just symptoms), generate possible solutions, evaluate and choose the best one, then implement and verify it worked.
- **Influence without Authority:** achieving cooperation from people who don't report to you by using relationship-building, expertise, credibility and persuasion instead of positional power.

10. COMMUNICATION

- **Communication Models:** a sender encodes a message and transmits it through a medium/channel; the receiver decodes it and (ideally) provides feedback; "noise" (distractions, language barriers, technical issues) can distort the message at any point.
- **Active Listening:** also central to good communication - confirming understanding through paraphrasing, asking clarifying questions, and paying attention to non-verbal cues.
- **Push Communication:** information sent directly to specific recipients who need to receive it (e.g. emails, memos, reports) - doesn't guarantee it was actually received or understood.
- **Pull Communication:** information is made available for recipients to access at their own discretion (e.g. intranet sites, knowledge repositories, shared drives) - efficient for large audiences with varied information needs.
- **Interactive Communication:** real-time, multidirectional exchange between two or more parties (meetings, phone calls, video conferences) - generally the most effective method for complex, sensitive or urgent information because it allows immediate feedback and clarification.

- Communication Barriers: noise, language differences, cultural differences, physical distance, distractions, poor listening skills, and unstated assumptions can all block effective communication.
- Stakeholder Communication: the frequency, level of detail, and format of communication should be tailored to each stakeholder's needs and influence, as documented in the Stakeholder Engagement Plan and Communications Management Plan.

11. RISK MANAGEMENT

- Positive Risks (Opportunities): risks that, if they occur, would have a beneficial effect on one or more project objectives - should be actively pursued.
- Negative Risks (Threats): risks that, if they occur, would have a harmful effect on one or more project objectives - should be actively managed down.
- Risk Appetite: the general degree of uncertainty an organization or stakeholder is willing to accept in anticipation of a reward; describes overall attitude toward risk-taking.
- Risk Threshold: the measure of acceptable variation around a specific objective that reflects the organization's/stakeholder's risk appetite (e.g. "we can tolerate up to a 10% budget overrun").
- Risk Tolerance: the degree, amount or volume of risk that an organization or individual will withstand for a particular risk category.
- Probability and Impact Matrix: a tool used to prioritize identified risks for further action by combining their likelihood of occurring with the severity of their effect if they do.
- Risk Response Strategies: for Threats - Avoid, Mitigate, Transfer, Accept, Escalate; for Opportunities - Exploit, Enhance, Share, Accept, Escalate (see full definitions in Section 5, Risk).

12. PROCUREMENT (KNOW ALL CONTRACT TYPES)

Fixed Price (FP) - seller bears most of the cost risk

- FFP (Firm Fixed Price): a single set price for defined work, regardless of the seller's actual costs; simplest and most common fixed price contract, best when requirements are well defined.
- FPIF (Fixed Price Incentive Fee): a fixed price plus a financial incentive tied to meeting agreed metrics (e.g. schedule or performance targets), with a price ceiling.
- FP-EPA (Fixed Price with Economic Price Adjustment): used for long-term contracts, allows for pre-defined final price adjustments due to changed conditions such as inflation or material cost changes.

Cost Reimbursable (CR) - buyer bears most of the cost risk

- CPFF (Cost Plus Fixed Fee): the seller is reimbursed for allowable costs plus receives a fixed fee agreed at contract signing, regardless of actual costs.
- CPIF (Cost Plus Incentive Fee): the seller is reimbursed for costs and receives a predetermined incentive fee based on achieving specific performance objectives.
- CPAF (Cost Plus Award Fee): the seller is reimbursed for costs and can earn a subjective award fee based on the buyer's satisfaction, per criteria in the contract.

Time & Material (T&M;)

- A hybrid arrangement combining aspects of fixed price and cost-reimbursable contracts; the buyer pays an agreed rate per hour/unit for labour plus reimbursement for materials, often used for staff augmentation or when the full scope isn't known upfront. Risk is shared between buyer and seller, since costs are somewhat open-ended but capped by rates.

13. QUALITY

- Prevention vs Inspection: Prevention means keeping errors out of the process in the first place (better, cheaper approach); Inspection means keeping errors out of the customer's hands by finding them after the work is done (catches defects late, more costly to fix).
- Cost of Quality: Cost of Conformance = Prevention costs (training, process documentation, equipment) + Appraisal costs (testing, inspections, destructive testing loss); Cost of Nonconformance = Internal Failure costs (rework, scrap) + External Failure costs (warranty claims, lost business, liabilities).
- Continuous Improvement (Kaizen): an ongoing, incremental effort to improve products, services or processes over time, rather than relying on occasional large changes.
- Root Cause Analysis: a technique used to identify the true underlying cause of a problem or defect, rather than just addressing its visible symptoms, so it doesn't recur.
- Pareto Principle/Diagram: based on the 80/20 rule (roughly 80% of problems come from 20% of causes); a Pareto diagram is a bar chart ranking causes by frequency to help teams focus improvement efforts on the "vital few".
- Fishbone (Ishikawa/Cause-and-Effect) Diagram: a visual tool that maps potential causes of a specific problem, often grouped into categories such as People, Process, Equipment, Materials, Environment.
- Control Charts: show whether a process's output is within acceptable statistical control limits over time, helping distinguish normal variation from a process going out of control.
- Histograms: bar charts showing the frequency distribution of numerical data, useful for spotting patterns like where most defects cluster.
- Scatter Diagrams: plot two variables against each other to visually show whether and how strongly they are correlated.

14. CHANGE MANAGEMENT

- Change Request: a formal proposal to modify any document, deliverable or baseline (scope, schedule, cost); must go through the change control process before being approved.
- Change Log: a document used to track the status of all change requests made during the project, including who submitted them and the final decision.
- Change Control Board (CCB): a formally chartered group responsible for reviewing, evaluating, approving, delaying or rejecting changes to the project, and for recording/communicating decisions.
- Configuration Management: ensures the description of the project's products is correct and complete; controls changes to the functional and physical characteristics/features of the product, distinct from (but related to) change control of the overall project.
- Baselines: the approved versions of scope, schedule and cost used as the basis for comparing actual performance; baselines can only be changed through the formal integrated change control process, never informally.

15. EARNED VALUE MANAGEMENT (VERY IMPORTANT)

Key Terms

- PV (Planned Value): the authorized budget assigned to scheduled work - i.e. how much work SHOULD have been done, in budget terms, by this point in time.
- EV (Earned Value): the value of the work actually completed, expressed in terms of the approved budget for that work - i.e. how much work HAS actually been done, in budget terms.
- AC (Actual Cost): the actual cost incurred for the work completed in a given period - i.e. how much has actually been SPENT.
- BAC (Budget at Completion): the total planned budget for the entire project.
- EAC (Estimate at Completion): the expected total cost of completing all project work, forecast using current performance.
- ETC (Estimate to Complete): the expected cost needed to complete all remaining project work from this point forward.
- VAC (Variance at Completion): the expected difference between the original budget (BAC) and the final forecasted cost (EAC).
- CPI (Cost Performance Index): a measure of the cost efficiency of budgeted resources.
- SPI (Schedule Performance Index): a measure of schedule efficiency, expressed as the ratio of earned value to planned value.
- CV (Cost Variance): the difference between earned value and actual cost, showing whether the project is under or over budget.
- SV (Schedule Variance): the difference between earned value and planned value, showing whether the project is ahead of or behind schedule.
- TCPI (To-Complete Performance Index): the projected cost performance that must be achieved on remaining work to meet a specified management goal (BAC or EAC).

Formulas

- $CV = EV - AC$ (Positive = under budget/good, Negative = over budget/bad, Zero = exactly on budget)
- $SV = EV - PV$ (Positive = ahead of schedule/good, Negative = behind schedule/bad, Zero = exactly on schedule)
- $CPI = EV / AC$ (greater than 1 = under budget, less than 1 = over budget, equal to 1 = on budget)
- $SPI = EV / PV$ (greater than 1 = ahead of schedule, less than 1 = behind schedule, equal to 1 = on schedule)
- EAC (variance is atypical, one-time) = $AC + (BAC - EV)$
- EAC (variance is typical and will continue at the same rate) = BAC / CPI
- EAC (both CPI and SPI factors will influence remaining work) = $AC + [(BAC - EV) / (CPI \times SPI)]$
- $ETC = EAC - AC$
- $VAC = BAC - EAC$
- $TCPI$ (goal is BAC) = $(BAC - EV) / (BAC - AC)$
- $TCPI$ (goal is a revised EAC) = $(BAC - EV) / (EAC - AC)$

Interpretation Tips

- CPI/SPI exactly 1 means the project is perfectly on target for cost/schedule respectively. Values greater than 1 are favourable (ahead of schedule / under budget); values less than 1 are unfavourable (behind schedule / over budget).
- When calculating EAC, first decide whether the variance is a one-time, unusual cost (use $AC + BAC - EV$) or reflects an ongoing trend that will continue (use BAC/CPI). Look for cues such as "the variance is typical of future variances" vs "this was an unusual one-time cost" to decide which formula applies.

16. PMP FORMULAS (QUICK REFERENCE)

- Communication Channels = $N(N-1)/2$, where N = number of stakeholders/team members; shows how communication paths increase rapidly as team size grows.
- PERT (Expected Duration) = $(Optimistic + 4 \times \text{Most Likely} + Pessimistic) / 6$; a weighted average that gives more weight to the most likely estimate.
- Standard Deviation = $(Pessimistic - Optimistic) / 6$; measures the amount of uncertainty/spread around the estimate.
- Variance = Standard Deviation squared; used in more advanced statistical risk calculations.
- $CPI = EV/AC$ | $SPI = EV/PV$ - the two key efficiency ratios for cost and schedule.
- $CV = EV - AC$ | $SV = EV - PV$ - the two key variance (difference) measures for cost and schedule.
- $VAC = BAC - EAC$ - shows the expected over/under run at project completion.
- $EAC = AC + (BAC - EV)$ [one-time variance] or BAC/CPI [variance continues] or $AC + (BAC - EV)/(CPI \times SPI)$ [both factors continue].
- $ETC = EAC - AC$ - what is still needed to finish the project from today.
- $TCPI = (BAC - EV)/(BAC - AC)$ - the efficiency needed on remaining work to hit the original budget.
- Float (Total Float) = Late Start - Early Start (or equivalently, Late Finish - Early Finish) for an activity.
- Free Float = the earliest start date of the next activity minus the earliest finish date of the current activity, without delaying any successor's early start.

17. ETHICS - PMI CODE OF ETHICS AND PROFESSIONAL CONDUCT

- Responsibility: taking ownership for the decisions we make or fail to make, the actions we take or fail to take, and the consequences that result; this includes owning up to and correcting our own mistakes, and holding ourselves accountable for any violations we have knowledge of.
- Respect: showing a high regard for ourselves, others, and the resources entrusted to us (people, money, reputation, the safety of others, natural/environmental resources); treating people and property with dignity, listening to other viewpoints, and approaching disagreements respectfully.
- Fairness: making decisions and acting impartially and objectively; avoiding favouritism, discrimination, and conflicts of interest, and proactively disclosing any real or potential conflicts to the appropriate stakeholders.
- Honesty: understanding the truth and acting in a truthful manner both in our communications and in our conduct; not engaging in or condoning behaviour designed to deceive others, such as making misleading statements, providing false information, or misrepresenting facts.

18. COMMON DOCUMENTS - PURPOSE AND CONTENTS

- **Project Charter:** the document that formally authorizes the project's existence and gives the PM authority to apply organizational resources; typically includes purpose, measurable objectives, high-level requirements/assumptions/risks, summary milestones, and the approval/sponsor signature.
- **Business Case:** the economic feasibility study that justifies the investment in the project before it formally begins; includes business need, options analysis, and recommendation.
- **Benefits Management Plan:** describes how and when the project's intended benefits will be delivered, and the mechanisms that should be in place to measure those benefits (including who is accountable after the project closes).
- **Project Management Plan:** the master document that integrates and consolidates all of the subsidiary management plans (scope, schedule, cost, quality, resource, communication, risk, procurement, stakeholder) and baselines; approved by the sponsor and used to guide project execution and control.
- **Scope Baseline:** the approved Scope Statement, WBS, and WBS Dictionary combined; the basis for comparing actual scope delivered against what was planned.
- **Schedule Baseline:** the approved version of the schedule model, used as the basis for comparing actual progress against the plan.
- **Cost Baseline:** the approved, time-phased budget, used as the basis for measuring and monitoring actual cost performance.
- **Risk Register:** the log listing all identified risks, their characteristics (category, root cause, owner, probability/impact) and their planned responses; updated continually.
- **Risk Report:** a higher-level summary document presenting overall project risk exposure along with the most significant individual risks, typically for sponsor/steering-committee-level reporting.
- **Issue Log:** a document where current issues (problems that have already occurred, as opposed to risks which are uncertain future events) are recorded, assigned an owner, and tracked until resolved.
- **Assumption Log:** records all assumptions and constraints identified throughout the project life cycle; assumptions later proven false can become risks, so this log should be reviewed regularly.
- **Lessons Learned Register:** captures knowledge gained during the project - both what worked well and what didn't - so that future projects can benefit and repeat mistakes can be avoided.
- **Stakeholder Register:** identifies project stakeholders and details their assessment and classification information (interests, involvement, impact, influence).
- **Change Log:** tracks the status of every change request submitted during the project, including the decision made and by whom.
- **Procurement Documents:** the collection of RFP/RFQ/RFI solicitation documents, along with contracts, statements of work, and other agreements with external sellers.
- **Work Performance Reports:** the physical or electronic representation of work performance information (status, progress, forecasts) compiled and formatted to support decisions, actions or awareness by stakeholders.
- **Final Report:** a summary of overall project performance delivered at project closure, covering scope, quality, cost, schedule performance, and validation of the final product against acceptance criteria, along with a summary of benefits achieved.

19. PROCESS GROUP MAPPING (HOW THE PROCESS GROUPS CONNECT)

- The five process groups are not strictly sequential phases - they overlap and interact throughout the project, but the general flow of authority and work is:
- Initiating -> Planning -> Executing -> Monitoring & Controlling -> Closing.
- Monitoring & Controlling runs in parallel with Planning, Executing and even Initiating/Closing - it is a continuous overlay checking performance and driving change requests back into Planning/Executing.
- A typical project has repeated Plan -> Execute -> Monitor/Control loops (especially across phases), not one single pass.

Where Each Major Process Belongs

Process Group	Key Processes
Initiating	Develop Project Charter; Identify Stakeholders
Planning	Develop PM Plan; Plan Scope/Schedule/Cost/Quality/Resource/Communication/Risk/Procurement/Stakeholder Management; Collect Requirements; Define Scope; Create WBS; Define & Sequence Activities; Estimate Durations & Costs; Develop Schedule; Determine Budget; Identify Risks; Perform Qualitative/Quantitative Risk Analysis; Plan Risk Responses
Executing	Direct and Manage Project Work; Manage Project Knowledge; Manage Quality; Acquire/Develop/Manage Resources; Manage Communications; Implement Risk Responses; Conduct Procurements; Manage Stakeholder Engagement
Monitoring & Controlling	Monitor and Control Project Work; Perform Integrated Change Control; Validate Scope; Control Scope/Schedule/Cost/Quality/Resources; Monitor Communications; Monitor Risks; Control Procurements; Monitor Stakeholder Engagement
Closing	Close Project or Phase

- Quick reference: work that authorizes a project or phase belongs to Initiating; preparing a plan/baseline belongs to Planning; doing the actual work belongs to Executing; comparing actual vs planned or approving changes belongs to Monitoring & Controlling; formal sign-off, handover and archiving belong to Closing.

20. ITTO AWARENESS (COMMON CROSS-CUTTING DOCUMENTS)

- You do not need to memorize the full Inputs/Tools & Techniques/Outputs list for every one of the 49 processes, but you should recognize these commonly recurring items and know roughly where each fits in the flow of a project:
- Project Charter: output of Develop Project Charter (Initiating); becomes a key input to almost all subsequent planning processes.
- Project Management Plan: the master output of Planning; becomes the primary input that guides Executing and Monitoring & Controlling.
- Work Performance Data: raw, unprocessed observations and measurements collected during Executing (e.g. number of change requests logged, actual start/finish dates, actual costs so far) - not yet analyzed.

- **Work Performance Information:** work performance data that has been analyzed and compared against the project baselines to give it context (e.g. calculated CPI/SPI values, percent complete versus planned) - produced during Monitoring & Controlling processes.
- **Work Performance Reports:** the physical or electronic compilation of work performance information, formatted into status reports, dashboards or memos for stakeholder decision-making and awareness - an output of the Monitor and Control Project Work process.
- **Change Requests:** formal proposals to modify any document, deliverable or baseline, generated whenever a variance or improvement opportunity is found; can come from any process.
- **Approved Change Requests:** change requests that have gone through Perform Integrated Change Control and been formally approved (typically by the CCB) - become an input to Direct and Manage Project Work for implementation.
- **Data flow direction:** Work Performance Data (raw) is turned into Work Performance Information (analyzed) which is compiled into Work Performance Reports (communicated).

21. AGILE MINDSET (PM DECISION-MAKING)

- A project manager's mindset and decision-making style matters as much as process knowledge, especially in agile and hybrid environments - the preferred approach reflects servant leadership and a team-empowering mindset rather than command-and-control.
- **Coach before directing:** help the team find their own solution through questions and guidance rather than telling them exactly what to do.
- **Facilitate instead of commanding:** guide discussions and decisions rather than dictating outcomes; let the team reach consensus where possible.
- **Empower the team:** trust the team (especially a self-organizing agile team) to make decisions about how the work gets done.
- **Remove blockers:** the PM's/Scrum Master's job is to clear obstacles so the team can focus on delivering value, not to micromanage tasks.
- **Escalate only when necessary:** try to resolve issues within the team first; escalate to the sponsor or management only when the issue is truly outside the team's authority or resources.
- **Deliver customer value continuously:** prioritize work and decisions based on what delivers the most value to the customer soonest, not just what completes the plan.
- When two options both seem procedurally correct, the one that reflects trust, empowerment, and servant leadership over direct control is generally preferred.

22. SITUATIONAL DECISION GUIDE (SCENARIO -> BEST FIRST ACTION)

Scenario	Best Sequence of Action
Customer requests a change	Assess the impact on scope/schedule/cost/risk; submit a formal change request through the change control process; do not implement informally unless it is a true emergency

Scenario	Best Sequence of Action
Team conflict	Let the team try to resolve it themselves first (self-organizing principle); if unresolved, facilitate using a collaborate/problem-solve approach; document and escalate only if it affects performance or safety
Risk occurs (becomes an issue)	Check the Risk Register for a planned response and implement it; if no response was planned, log it in the Issue Log, analyze root cause, then determine and implement an appropriate response
Stakeholder is unhappy	Meet with the stakeholder to actively listen and understand their concern; update the Stakeholder Engagement Plan/approach as needed; do not ignore or escalate before trying to understand the issue
Scope increases (scope creep)	Compare against the Scope Baseline; if it is truly new/changed scope, route it through integrated change control via a formal change request before proceeding
Budget overrun	Analyze the root cause using EVM data (CPI, CV); inform the sponsor with data-backed analysis; consider corrective action or use of contingency/management reserve as appropriate
Vendor delay	Review the contract terms and communicate with the vendor to understand the cause; assess impact on the project schedule; escalate per the contract's remedy clauses only if informal resolution fails

- General approach: identify the actual root problem first, follow the defined process (don't skip steps or act unilaterally), communicate with the people affected, and only escalate when the resolution is genuinely outside the PM's authority.

23. AGILE VS PREDICTIVE COMPARISON TABLE

Aspect	Predictive	Agile
Scope	Fixed scope, defined upfront	Flexible scope, evolves over time
Planning	Heavy upfront planning	Adaptive, just-in-time planning
Change	Formal change control process	Continuous reprioritization of the backlog
Delivery	Sequential phases, one delivery at the end	Iterative increments, frequent working deliverables
Customer involvement	Mainly at milestones/gate reviews	Continuous, throughout every iteration
Requirements	Detailed and documented early	High-level initially, elaborated progressively
Risk approach	Addressed mainly during planning	Addressed continuously each iteration
Best suited for	Well-understood, stable requirements	Uncertain, evolving requirements
Progress measured by	Earned Value (CPI/SPI), % complete against baseline	Velocity, burndown/burnup, working increments

24. RISK RESPONSE QUICK TABLE

Risk Type	Response Strategies (in typical order of preference)
Threats (negative risks)	Avoid - eliminate the threat entirely; Mitigate - reduce probability/impact; Transfer - shift ownership (e.g. insurance, outsourcing); Accept - acknowledge and do nothing or reserve for it; Escalate - pass to a level outside the project's authority
Opportunities (positive risks)	Exploit - ensure the opportunity definitely happens; Enhance - increase its probability/impact; Share - partner with a third party better able to capture it; Accept - take advantage if it happens but don't actively pursue it; Escalate - pass to a level outside the project's authority

- Threats and Opportunities are mirror images of each other - Avoid/Exploit are the strongest actions, Mitigate/Enhance adjust the odds, Transfer/Share involve a third party, and Accept/Escalate are shared by both.

25. STAKEHOLDER POWER/INTEREST GRID

Power / Interest	Strategy
High Power / High Interest	Manage Closely - engage frequently, involve in key decisions
High Power / Low Interest	Keep Satisfied - meet their needs, don't over-communicate
Low Power / High Interest	Keep Informed - provide regular updates, address concerns
Low Power / Low Interest	Monitor - minimal effort, watch for changes in power/interest

- Stakeholder classification is not permanent - power and interest can shift as the project progresses, so the grid should be revisited regularly, not just created once at the start.

26. CONFLICT RESOLUTION PRIORITY (MEMORABLE ORDER)

- A simple order to remember, from most to least preferred:
- 1. Collaborate / Problem-Solve (best) - works together to find a true win-win solution; best for important, complex issues where the relationship and outcome both matter.
- 2. Compromise - each side gives up something to reach a mutually acceptable middle ground; a lose-lose in the sense that neither side gets everything, used when time is limited.
- 3. Smooth / Accommodate - emphasizes areas of agreement over differences to preserve the relationship; suitable for minor issues or when preserving harmony matters more than winning the point.
- 4. Force / Direct - one party pushes their viewpoint through at the expense of others; a win-lose approach, appropriate mainly in urgent situations or when a decision must be made quickly.

- 5. Withdraw / Avoid (least preferred) - retreating from or postponing the conflict; only appropriate for trivial issues or to allow tempers to cool before addressing the real issue later.
- Collaborate is generally the best approach when the scenario allows time for it; Withdraw/Avoid and Force/Direct are rarely the ideal choice unless the situation specifically calls for urgency or the issue is trivial.

27. SERVANT LEADERSHIP SUMMARY

- Definition: a leadership philosophy where the leader's primary role is to serve the team - removing obstacles, providing resources, and empowering people to do their best work - rather than directing and controlling from the top down.
- Contrast with traditional leadership: traditional/command-and-control leadership focuses on the leader directing tasks and making most decisions; servant leadership flips the pyramid so the leader supports the team, and decision-making authority is pushed down to those doing the work.
- Core responsibilities of a servant leader:
 - Remove impediments and blockers that slow the team down.
 - Protect the team from unnecessary distractions and outside interference so they can focus on delivering value.
 - Coach and mentor team members to help them grow their skills and confidence.
 - Facilitate discussions and decision-making rather than dictating outcomes.
 - Promote self-organization - trust the team to determine the best way to do the work.
 - Build a safe, collaborative environment where people feel comfortable raising problems early.
 - Champion the team's needs to the rest of the organization, and shield them from unreasonable external pressure.
- Why it matters: agile and hybrid ways of working place heavy weight on the PM's leadership mindset, and the preferred response is generally the servant-leadership one (coach, empower, remove blockers, facilitate) over the traditional command-and-control response (direct, assign, decide alone).

28. WHAT I LEARNED

- The **foundations of project management**: what a project is, how it differs from ongoing operations, the constraints that shape every project, and how project managers are positioned within different organizational structures.
- The **business context** projects operate in: business value, benefits management, governance, compliance, and how projects connect to broader organizational strategy through the value delivery system.
- The **12 guiding principles** and **8 performance domains** that describe how a project manager should think and operate across any type of project, regardless of development approach.
- The full **predictive (Waterfall) process flow** across all five process groups and ten knowledge areas, including the core documents, techniques and controls used in each.
- The **Agile framework** in depth: its values and principles, common frameworks (Scrum, Kanban, Lean, XP, Crystal, DSDM, FDD), Scrum roles/artifacts/events, and how agile teams estimate, measure progress and maintain quality.
- How **hybrid approaches** combine predictive and agile elements, and when each is appropriate.

- The **soft skills** that underpin effective project leadership: emotional intelligence, conflict resolution, negotiation, motivation, coaching, mentoring and communication.
- Core **risk, procurement, quality and change management** concepts, along with the full set of **Earned Value Management and PMP formulas**.
- The **PMI Code of Ethics** and the purpose of the key documents used throughout a project's life cycle.
- How the process groups connect end to end, common cross-cutting documents (Work Performance Data/Information/Reports, Change Requests), the **agile mindset** a project manager should bring to decisions, and quick-reference tools such as the comparison table, risk response table, stakeholder grid, and conflict resolution order.

29. MAIN TAKEAWAYS

- **Value**, not just scope/time/cost, is the ultimate measure of project success - a project can hit its constraints and still fail to deliver what the organization actually needed.
- **Tailoring the approach** (predictive, agile or hybrid) to the project's context matters more than rigidly following one methodology.
- **Stakeholder engagement and communication** are continuous activities, not one-time events - both should be revisited throughout the project as needs and influence change.
- **Risk management is proactive, not reactive** - responses should be planned before risks occur, for both threats and opportunities.
- Changes to any baseline must go through **formal, integrated change control** - informal or verbal changes put the project at risk.
- **Earned Value Management formulas** (CPI, SPI, CV, SV, EAC, TCPI) are essential tools for objectively measuring and forecasting project performance.
- Agile and hybrid environments call for a **servant-leadership mindset** - coaching, facilitating and empowering the team rather than directing and controlling.
- Conflict, negotiation and decision-making all have a **preferred order of approach** - collaboration and root-cause thinking generally lead to better outcomes than avoidance or force.
- Every knowledge area and performance domain connects back to the same goal: **delivering the intended benefits** to the organization and its stakeholders.